A fusion-junction vaccine in extraskeletal myxoid chondrosarcoma: what can be established today, and the capabilities that would change it

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Running title. A junction vaccine in EMC: what is established

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Abstract

Extraskeletal myxoid chondrosarcoma (EMC), a rare cancer, usually carries a join between the *EWSR1* and *NR4A3* genes. A vaccine might train T cells to recognize short protein fragments spanning that join. We asked what current computational methods can establish about this approach and what still requires experiments.

We reconstructed fusion sequences, screened their fragments and checked for matches in normal proteins. Human leukocyte antigen (HLA) proteins display fragments to T cells, and their inherited variants differ between people. We therefore estimated how many people carry an HLA variant predicted to display at least one candidate.

Four candidate fragments also occurred in an annotated normal *NR4A3* protein variant. Correcting this missed match removed one predicted binder; all four fragments classified as strong binders remained. A leading candidate differed from a normal protein fragment by just one amino acid. Whether T cells recognize it or cross-react with normal tissue is unknown.

Predicted population coverage changed from 0% to 72.6% when the acceptance threshold changed on the wider HLA panel. It also changed with the variants screened, prediction method and population. These figures describe model-dependent HLA carriage, not the proportion of patients whose tumors display a candidate or respond to vaccination.

The study identifies a failure in the original normal-protein check and shows why predicted coverage is not a biological limit. No laboratory data were generated. Actual tumor display, immune recognition, safety and benefit remain unmeasured; confirming the fusion sequence and testing those biological steps must precede a judgment of vaccine feasibility.

1. A standing-state report rather than a verdict

This paper asks which obstacles to an EMC fusion-junction vaccine could change with better evidence. A negative prediction alone cannot tell us whether the target is unsuitable or whether the available method cannot assess it. We therefore distinguish three kinds of limitation. Disease-related limits include a quiet genome, an incidence below one per million per year and a myxoid matrix proposed to exclude lymphocytes. Method-related limits arise when sequence predictions substitute for measurements. Access-related limits include unavailable patient material, unmeasured presentation and the lack of a manufacturing route at this incidence.

Section 3 identifies the evidence that would change each movable limitation. It gives no arrival dates. Section 6 explains how to distinguish that evidence from a similar-looking result that would leave the limitation unresolved.

The immediate occasion is external. The sponsors of an individualised neoantigen therapy given with pembrolizumab have announced that a randomised phase 3 trial met its primary endpoint of recurrence-free survival in resected stage IIB to IV melanoma [9]. The announcement is a company press release and discloses no effect size. The peer-reviewed evidence in this setting remains the phase 2b trial [3]. The result does not transfer to EMC, and Section 4 sets out the axis on which the transfer fails. What it shows is that the manufacturing and delivery apparatus for an individualised RNA vaccine exists as a clinical reality rather than as a proposal.

A closer precedent than melanoma exists for the modality proposed here, and it is much smaller. A 9-mer peptide spanning the SYT-SSX junction of synovial sarcoma, restricted to HLA-A*24:02, was given to six patients in a phase I trial [17] and then to 21 patients across four protocols by the Japanese Musculoskeletal Oncology Group [18]. Peptide-specific cytotoxic T cells were induced in some patients, and delayed-type hypersensitivity responses were absent in both trials. Half the 12 patients on the arms combining peptide with incomplete Freund's adjuvant and interferon-alpha had stable disease during the vaccination period, and one patient in that series developed an intracerebral haemorrhage after the second vaccination. An accompanying evaluation of the later trial concluded that no robust immune response to the target epitope had been shown [19]. More recently, an off-the-shelf multi-peptide vaccine spanning the type 1 *EWSR1*::*FLI1* breakpoint, given with GM-CSF and topical imiquimod after multimodal therapy, raised de novo polyfunctional CD4+ T-cell responses against all four fusion-derived peptides in one patient with metastatic Ewing sarcoma, first detected at about month 7 and persisting beyond two years [20]. That report is a single patient and carries no control arm, so it supports no response rate and no statement about how such a construct would perform in anyone else. Fusion-derived neoantigens have separately been shown to elicit cytotoxic T-cell responses in fusion-driven tumours, alongside evidence of negative selective pressure against them [21]. The human evidence for fusion-junction vaccination is therefore one case report and two small early-phase trials in a different translocation sarcoma: more than nothing, and much less than a demonstration that the approach works. Nothing in it concerns *EWSR1*::*NR4A3* or EMC.

2. Results: the target and its current evidence base

2.1 Disease and fusion

EMC accounts for roughly 1 to 3% of soft-tissue sarcomas, with an estimated incidence well under one per million per year [7]. It is defined by rearrangement of *NR4A3* on chromosome 9q22. *EWSR1*::*NR4A3* is the commonest fusion, reported in approximately 62 to 75% of cases [7] and in 79% of a molecularly confirmed series of 58 cases [10]; variant partners include *TAF15*, *TCF12*, *TFG* and *FUS* [7]. The genome is otherwise quiet, and the fusion is the truncal driver.

Two consequences follow, and they point in opposite directions. Because the fusion is truncal and clonal, it is present in every tumour cell and cannot be subclonally lost, so a T-cell response directed at the junction cannot be escaped by antigen loss in the way that a response against a passenger mutation can. Because the genome is quiet, the junction is close to the only tumour-exclusive antigen the disease offers, so if the junction fails there is no second candidate to fall back on. The same feature that makes the target durable makes the portfolio of targets shallow.

2.2 Junction structure and predicted binding

A *peptide* is a short amino-acid sequence. An *epitope* is the part recognized by an immune receptor. HLA class I and class II presentation concern the CD8 and CD4 T-cell arms, respectively. In this paper, a predicted binder is a computational candidate: binding, tumour-cell presentation and T-cell recognition are distinct questions.

The transcript model retains sequence that a coding-sequence-only model would discard. We derived junctions from spliced transcripts and retained the whole acceptor exon, including its 5' untranslated region. A superseded model joined coding sequences alone. It removed that retained region and produced a different junction. All figures below use the transcript model.

Here, the *donor* supplies the first part of the fusion transcript and the *acceptor* supplies the second. The *seam* is the sequence boundary between them. An *in-frame* junction preserves the codon grouping needed to encode the intended fusion protein; an out-of-frame junction changes that grouping.

Of 27 declared exon pairs, 5 are in frame: *EWSR1* exons 7, 9, 10, 12 and 13 joined to *NR4A3* exon 3. The remaining 22 are graded out as non-coding acceptor (9), out of frame (4) or not producing the seam (9). The in-frame set yields 174 distinct junction-spanning peptides, screened with MHCflurry 2.1.4, models release 2.2.0 [2], over ten alleles — HLA-A*01:01, A*02:01, A*03:01, A*11:01, A*24:02, B*07:02, B*08:01, B*15:01, B*35:01 and B*44:02 — calling a peptide strong at a presentation percentile of 0.5 or below and weak at 2.0 or below. That screen returns 11 distinct predicted binders, 4 of them strong. The lead candidate at the commonly reported *EWSR1* exon 7 junction is NMPCVQAQY on HLA-B*15:01, at a presentation percentile of 0.37 and a predicted affinity of 73.4 nM. The class call is made on the percentile, not the affinity, and the two do not agree in rank order across this set, so affinities appear below only beside the percentile that classified them.

The following steps define the junction model and the original novelty filter. Section B5 identifies the filter's failure. Stating the steps makes clear where that failure occurs.

1. Build the chimeric mRNA as the donor transcript through the 3' end of exon *d*, then acceptor exon *a* **whole** — its 5' untranslated region included, as a fusion transcript retains it — and everything 3' of it.
2. Translate from the donor's own initiator codon, so the reading frame is the donor's throughout.
3. Let j_0 be the index of the first residue not encoded wholly by donor nucleotides. If the donor cut leaves a partial codon, the residue at j_0 is completed by acceptor nucleotides and belongs to neither parent; if the cut is codon-aligned there is no such residue.
4. Verify the seam rather than assume it: the chimeric N-terminus must match the donor protein residue for residue up to j_0, the C-terminus must be the acceptor's own, and acceptor residue 1 must appear

immediately 3' of the seam codon. A junction failing any of the three is an error, not a result.

5. Enumerate every 8- to 11-mer containing j_0 — including those that begin at it, which a straddle test requiring a residue on each side would drop.
6. **The novelty filter, and the step Section B5 is about:** keep a peptide if it is not a substring of wild-type *EWSR1* and not a substring of wild-type *NR4A3*. That is a two-protein test. It does not consult the rest of the proteome, and it does not consult other isoforms of either parent — which is precisely how four peptides that occur in a normal *NR4A3* isoform passed it.

Step 6 tested only two canonical protein sequences. It could therefore miss a match in another protein or in another isoform of either parent. Figure 1 shows the resulting seam and an out-of-frame example. Table 4 lists all 27 exon pairs and their grades, including the reasons 22 pairs were not carried forward.

Table 4. Every candidate exon pair, and why each is or is not carried forward. All 27 pairs the declared window produces, graded by the screen of §2.2: 5 in frame, 4 out of frame, 9 acceptor exon is non-coding, 9 no seam produced at this pair. *Donor phase* is the number of coding nucleotides of the donor's last codon left over at the cut, so phase 0 is a codon-aligned break and phases 1 and 2 build the seam codon from retained acceptor nucleotides. An *In frame* cell reading n/a is a pair with no chimeric reading frame for the question to be about, not a pair that failed it. These are a combinatorial window of declared exon pairs, not observed breakpoints, and the window is an assumption of this work rather than a finding of it (§2.2).

Table 4

EWSR1 exon	*NR4A3* exon	Donor phase	In frame	Grade
7	3	1	yes	in frame; peptides emitted
9	3	1	yes	in frame; peptides emitted
10	3	1	yes	in frame; peptides emitted
12	3	1	yes	in frame; peptides emitted
13	3	1	yes	in frame; peptides emitted
6	3	2	no	out of frame; read-through tract, then a premature stop
8	3	2	no	out of frame; read-through tract, then a premature stop
11	3	0	no	out of frame; read-through tract, then a premature stop
14	3	2	no	out of frame; read-through tract, then a premature stop
6	2	2	n/a	acceptor exon is non-coding; no chimeric ORF
7	2	1	n/a	acceptor exon is non-coding; no chimeric ORF
8	2	2	n/a	acceptor exon is non-coding; no chimeric ORF
9	2	1	n/a	acceptor exon is non-coding; no chimeric ORF
10	2	1	n/a	acceptor exon is non-coding; no chimeric ORF
11	2	0	n/a	acceptor exon is non-coding; no chimeric ORF
12	2	1	n/a	acceptor exon is non-coding; no chimeric ORF
13	2	1	n/a	acceptor exon is non-coding; no chimeric ORF
14	2	2	n/a	acceptor exon is non-coding; no chimeric ORF
6	4	2	n/a	no seam produced at this pair
7	4	1	n/a	no seam produced at this pair
8	4	2	n/a	no seam produced at this pair
9	4	1	n/a	no seam produced at this pair
10	4	1	n/a	no seam produced at this pair
11	4	0	n/a	no seam produced at this pair
12	4	1	n/a	no seam produced at this pair
13	4	1	n/a	no seam produced at this pair
14	4	2	n/a	no seam produced at this pair

That ten-allele panel is the instrument behind every binder figure here, and a wider one changes them: a 34-allele screen of the same peptides at the same threshold returns five strong peptide-allele calls rather than four, because NMPCVQAQY is also strong on HLA-A*30:02. Section 2.3 reports both panels.

The out-of-frame junctions. Twenty-two of the 27 pairs carry no peptides above, and four of those are out of frame — *EWSR1* exons 6, 8, 11 and 14 joined to *NR4A3* exon 3. A frameshifted junction reads the acceptor exon in a novel register, so every residue after the seam is non-self until a stop codon, and in other tumours that class is the richest antigen source available rather than the poorest. Here it is not. The four read-through tracts are 9, 9, 31 and 9 residues long before a premature stop, yielding 97 distinct 8- to 11-mers in total. Three of the four — those from *EWSR1* exons 6, 8 and 14 — converge on the identical eight-residue core YALRPSPI, differing only in the seam residue, because a frameshift into the same acceptor exon reads the same nucleotides in the same shifted register.

The antigen supply from this class is therefore one short peptide and one 31-residue tract, not four independent ones.

All four premature stops lie between 1,542 and 1,610 nucleotides upstream of the chimera's last exon-exon junction, which is the canonical configuration for nonsense-mediated decay by the 50-nucleotide rule — a positional criterion computed from the transcript model, not a decay measurement, and reported as such.

Screened against the same 34-allele panel at the same cut, those 97 peptides return 10 strong peptide-allele calls across 8 alleles, all of them from the single 31-residue tract, and the three 9-residue tracts return none.

Those calls are tighter than anything the in-frame junctions produce. The strongest is LPLQVPVM on HLA-B*51:01 at a presentation percentile of 0.1039, against a best in-frame call of 0.3736; four of the ten fall below 0.2755. Section 2.3 reports that a conservative 0.2 cut leaves the in-frame screen with no presenting allele at all — and the peptides that survive that cut anywhere in this work come exclusively from a junction that cannot produce the driver. The antigen-poor part of this locus is the part that makes the disease.

Two things bound this paragraph and are stated rather than implied: the 27 pairs are a combinatorial window of nine donor exons against three acceptor exons and not a set of observed breakpoints, and a frameshifted *EWSR1*::*NR4A3* cannot encode the chimeric transcription factor that defines the disease. So these peptides are not offered as EMC targets. What the screen establishes is the size of the antigen supply a frameshift at this locus would provide if one were ever observed, which is a property of the locus and answers the question rather than deferring it.

There is no pan-EMC epitope: the most widely shared candidate appears in 4 of the 5 junctions and is a weak binder, three of the five junctions return no strong binder at all, and every strong binder is specific to its breakpoint. One of the 11 predicted binders, DMPCVQAQY on HLA-B*35:01, is withdrawn on the proteome search reported under B5 below, leaving 10; all 4 strong binders survive that search.

2.3 Population coverage

Predicted coverage is the fraction of people expected to carry at least one HLA allele that the screen predicts can present a junction peptide. It is not the fraction whose tumours have been shown to present that peptide. Let A denote those predicted presenting alleles and f_a the pooled frequency of allele a in Allele Frequency Net Database records [1]. We calculate:

$$\mathbf{C(A) = 1 - \prod_{a \in A} (1 - f_a)^2}$$

Each factor $(1 - f_a)^2$ is the probability that an individual carries neither copy of allele *a* under Hardy-Weinberg, and the product treats loci as independent.

Every coverage figure in the running text uses f_a pooled across all AFND populations; Table 1 applies the same formula with f_a restricted to one sub-region's records, which is the only difference between its cells and the global row; *C* is therefore the fraction carrying at least one presenting allele.

Because *A* is itself a function of the acceptance threshold *t* — an allele enters *A* at the percentile of its best junction peptide — coverage is a function of *t*, and it is a right-continuous step function whose jumps are exactly the distinct peptide-allele percentiles. Section 2.3 computes it.

Three properties of that quantity govern how the figures below should be read: the panel, the threshold and the predictor. All three are properties of the screen rather than of the tumour. A fourth, below them, is a property of the population the figure averages over, and a fifth is the uncertainty the pooled figure carries and does not print.

It moves with the panel. On the ten-allele screen the *EWSR1* exon 7 junction is presented on HLA-B*15:01 alone and covers 8.5%; on 34 alleles the same lead peptide is also strong on HLA-A*30:02 and the junction covers 12.3%. Pooling every in-frame junction gives 27.4% on ten alleles and 30.4% on 34. The four figures are one computation at two panel widths and two junction scopes, not four findings, and the 27.4% and 30.4% pair differs by exactly one allele. Extending the panel further can only raise them, so none of them is a ceiling and this paper does not call any of them one.

It moves with the threshold further than it moves with anything else, and the whole function is computed here rather than sampled at three points. Coverage against the acceptance threshold is a step function whose every step is a single peptide-allele call. Below the conventional cut it has four. HLA-B*15:01 enters at presentation percentile 0.3736 on NMPCVQAQY, taking coverage from zero to 8.5%; HLA-A*30:02 at 0.4033 on the same peptide, reaching 12.3%; HLA-A*01:01 at 0.4061 on RGDMPCVQAQY, reaching 23.2%; HLA-B*07:02 at 0.4580 on MPPPLRGDM, reaching 30.4%.

Four steps and five strong calls, because the fifth — QQNMPCVQAQY on HLA-B*15:01 at 0.4986 — adds a second peptide to an allele already presenting and moves nothing.

Sampling at round numbers, as earlier versions of this paper did — to 0.45 leaves three alleles and 23.2%, to 0.40 leaves one and 8.5%, to 0.37 leaves none and 0.0% — understates how compressed the function is.

Every step falls inside a window 0.0844 units wide; the function is then flat from 0.4580 all the way to the cut; and it reaches zero 0.1264 below it. The headline figure is flat at the threshold not because the threshold is well placed but because it sits past the last step, with a cliff a tenth of a percentile unit beneath.

Above the cut it does not plateau either, and that is the more useful half. Extending the same screen to a percentile of 5 adds twenty-four further alleles in twenty-four further steps, none of them large and none of them absent: 8 presenting alleles and 47.8% at a cut of 1.0, 16 and 72.6% at 2.0, 28 and 90.2% at 5.0. The paper's own weak-binder cut is 2.0. So over the span of cuts this field routinely writes down — 0.2 as a conservative tier, 0.5 as strong, 2.0 as weak — predicted coverage of this junction runs from 0% to 72.6%, and the reported 30.4% is one arbitrary point inside that range rather than an estimate with a tolerance.

And it moves with the predictor, which is a third axis and was untested until now. Every figure above comes from one software suite, so the sensitivity reported here could be a property of that suite rather than of the junction.

The same 174 peptides and the same 34 alleles were therefore re-screened with MHCnuggets [12], an independently trained class I predictor whose native output is a predicted IC50 rather than a presentation percentile.

The two scales are not rescaled onto one axis — that would manufacture the agreement it reports — so each predictor is swept over its own conventional cut and the shapes are compared. Every allele on the panel returned a score, so no cell of the comparison is an absent reading.

Two things follow. The threshold sensitivity is not an artifact of one suite: the second predictor's curve is equally steep, running from no presenting allele at 10 nM through 8 alleles and 45.2% at its conventional 500 nM to 32 alleles and 94.0% at 10,000 nM.

But the two predictors do not agree on which alleles present: at each one's own conventional cut they share three — HLA-A*30:02, HLA-B*07:02 and HLA-B*15:01 — while HLA-A*01:01 is MHCflurry's alone and five more are MHCnuggets' alone.

So the choice of predictor moves the headline figure from 30.4% to 45.2% on identical inputs, and the intersection of the two is three alleles of a union of nine. The lead allele HLA-B*15:01 is in the intersection, which is the one place in this paper where two independent instruments say the same thing.

None of this is an argument for a looser cut. A threshold chosen because it raises coverage would be the same defect this paper exists to name, arriving from the other side. The 28 alleles at a percentile of 5 are not a better answer than the 4 at 0.5. They are a demonstration that the question "what fraction of patients could this reach?" has no answer until somebody defends a cut, and nobody has. Figure 2 plots the function on a log axis, which is the axis its shape needs: the four steps span 0.0844 percentile units and a linear axis collapses them into the left margin. The whole curve is deposited as `coverage-threshold-curve.json`, so a reader who would set the cut elsewhere can read off what it gives instead of taking three points on trust.

It is an average over populations that do not resemble one another. The pooled figure is a single number laid over a distribution whose ends are more than forty-fold apart, and a reader planning anything for a particular place needs the end they are standing at rather than the mean.

Predicted coverage of any in-frame junction runs from 1.4% in Melanesia to 60.4% in Northern Europe, and of the public *EWSR1* exon 7 junction from 0.8% in Northern Africa to 16.1% in Northern Europe.

So the statement that no panel screened here reaches half of patients is true of the pooled global frame and false in Northern Europe, where three alleles reach three fifths. This is not a limit of the screen — it is what HLA frequency is — but it means the pooled figure should not be quoted alone, and the table below is given so that it need not be.

Table 1. Predicted class I coverage by UN M49 sub-region, ten-allele screen. Coverage is the union carrier frequency of the presenting alleles over Allele Frequency Net Database records [1], computed within each sub-region rather than pooled. The pooled global figures are the last row. *Public junction* is the *EWSR1* exon 7 to *NR4A3* exon 3 junction, presented on HLA-B*15:01 alone; *any junction* pools every in-frame junction, presented on 3 alleles. No confidence interval is given, for the reason in §2.3; *n* is the largest per-allele sample the sub-region contributes and is what a reader should weigh a cell against. A dash is an allele absent from that sub-region's records, not a coverage of zero.

Table 1

Sub-region	Public junction	Any junction	Presenting alleles	*n*
Northern Europe	16.1%	60.4%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	1598
Western Europe	8.2%	41.6%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	4045
Eastern Europe	7.1%	37.0%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	2456
Southern Europe	5.9%	32.1%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	5982
Northern America	8.0%	31.6%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	5478
Western Asia	2.6%	27.8%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	2543
Southern Asia	4.0%	25.1%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	3580
Australia and New Zealand	5.0%	23.9%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	702
Northern Africa	0.8%	21.2%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	1537
Latin America and the Caribbean	7.8%	21.1%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	17277
Sub-Saharan Africa	2.0%	20.4%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	3066
Eastern Asia	15.4%	20.0%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	11585
South-eastern Asia	3.7%	14.0%	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	3556
Polynesia	—	1.9%	HLA-B*07:02	450
Melanesia	0.9%	1.4%	HLA-B*07:02, HLA-B*15:01	1269
Micronesia	—	—	—	129
Pooled global	**8.5%**	**27.4%**	HLA-A*01:01, HLA-B*07:02, HLA-B*15:01	—

What the withdrawn interval was, and what stands in its place. Earlier versions printed Wilson 95% intervals about half a percentage point wide on every figure above. They are withdrawn. They were computed by pooling every reference population into one binomial model. Yet the same records show HLA-B*15:01 frequencies ranging from 0 to 0.40 between populations. Those differences contradict the single-population sampling assumption. The interval is also an order of magnitude narrower than the threshold sensitivity above. The regional figures in Table 1 are point values for the same reason, and the reason is stronger there: the sample behind a sub-region's cell is smaller and less homogeneous than the pooled one, not more.

We replace the withdrawn interval with three complementary analyses. They examine the same-locus approximation, dependence between loci and variation between source populations. Each answers a different question. None is a confidence interval, and all three are recorded in `coverage-uncertainty.json`.

The same-locus approximation, and its direction. The formula above multiplies $(1 - f_a)^2$ over alleles, taking HLA-B*07:02 and HLA-B*15:01 to be independent events when a diploid genome carries at most two alleles at HLA-B and the two compete for the same pair of slots.

Grouping by locus is exact under the same Hardy-Weinberg assumption: the probability of carrying neither of a locus's presenting alleles is $(1 - \Sigma f_a)^2$, not $\prod (1 - f_a)^2$. Since $(1 - f_1 - f_2) < (1 - f_1)(1 - f_2)$, the published form is a lower bound and not an approximation of unknown sign.

Recomputed exactly, the pooled ten-allele figure is 27.7% against the 27.4% printed above: 0.33 percentage points, in the direction that means no coverage figure in this paper is too high for this reason.

The correction is small here only because the two same-locus frequencies are small; it grows with the panel, and a screen reporting many alleles at one locus under the published form would understate by considerably more.

Dependence between loci, bounded rather than assumed away. What remains is independence between HLA-A and HLA-B, which linkage disequilibrium violates and which nothing in this work measures. It does not have to be measured to be bounded. Pooled per-locus carrier frequencies are 12.4% at A and 17.5% at B, and for a union of events with those marginals the Fréchet bounds place coverage in [17.5%, 29.9%] under *any* dependence structure whatever, exactly and distribution-free, assuming nothing about haplotypes. Positive disequilibrium, presenting alleles clustering in the same people, pushes coverage toward the lower end; negative disequilibrium toward the upper. So the independence assumption is worth at most 12.4 percentage points here, and the reader who distrusts it can take the bound instead of the estimate.

Hardy-Weinberg is the assumption underneath that one, and it is bounded the same way. Reading a carrier frequency off an allele frequency through $(1 - p)^2$ assumes Hardy-Weinberg at the locus, and nothing here tests it: AFND publishes allele frequencies rather than genotype counts, so it cannot be tested from this source.

It can be bounded from it. An allele frequency is a fraction of chromosomes, and every way of arranging those copies into people lies between two extremes — every copy paired in a homozygote, giving a carrier frequency equal to p, and no homozygotes at all, giving $\min(1, 2p)$. Hardy-Weinberg's $2p - p^2$ sits inside that interval for every p, near its top when p is small.

Dropping Hardy-Weinberg and between-locus independence together, coverage of the ten-allele screen lies in [9.2%, 31.1%], and the 27.7% above sits inside it. That interval assumes nothing about genotypes or haplotypes at all, and it is the widest statement this paper can make that is still certainly true.

The spread that matters is between populations, and it is far wider than either. Recomputing C inside each AFND study population from that population's own frequencies, restricted to the 112 populations of at least 50 individuals that measured all three presenting alleles, gives a median of 24.5%, an interquartile range of 9.6% to 35.6%, and a full range of 0% to 66.0%.

The pooled 27.4% sits near the median of a distribution whose middle half spans 26 percentage points and whose extremes are 0% and 66.0%. That is an order of magnitude wider than either reading above it, wider than the panel and predictor axes of this same figure, and narrower only than the threshold axis.

Taken instead over every population with any measurement, scoring an unmeasured allele as absent, the median falls to 3.1% across 482 populations. That is a floor rather than an estimate, because absence from AFND is a reporting gap and not a measured zero.

The distance between those two readings is a statement about the database, not about HLA, and it is given here so that the 112-population figure is not read as a survey of the world.

None of the three is the quantity a confidence interval would give, and none is offered as one. What is still absent is any interval on the *screen* — on whether the peptide-allele calls that define A are correct. That is limit B2, which no rearrangement of population statistics can reach.

Because an individualised platform selects against the patient's own genotype rather than a public epitope, the relevant figure is the pooled-junction one. Selecting per patient moves it from 8.5% to 30.4% on the panels screened here, and removes neither the panel dependence nor the threshold dependence.

2.4 Limits of the current evidence

Predicted binding is a screen. It is not evidence that any peptide is presented on an EMC tumour cell, not a measure of the density at which it would be presented, and not evidence of T-cell recognition. No EMC immunopeptidomic dataset is known to the author. Sequence-level novelty against the proteome has now been tested and is reported under B5, which excludes one failure mode and leaves presentation, immunogenicity and cross-reactivity untouched. The machinery a presented peptide would require has since been read at the transcript level in EMC tumour tissue and is not lost (B8), which establishes a precondition rather than a presentation. These gaps are enumerated as limits B2 and B3 rather than treated as resolved.

3. The standing state, limit by limit

The table below organizes ten limitations by what determines them. Disease-bounded limits arise from the tumour and are not expected to move. Instrument-bounded limits may change when methods improve. Access-bounded limits reflect this programme's access to material or facilities, rather than a lack of knowledge in the field.

The cost-to-move column's units are whose permission and whose material a row requires, rather than money or months, neither of which this programme can estimate. No time estimate is offered for any row, for the reason Section 3.1 gives.

ID	Limit	Bounded by	Best available answer today	Cost to move	What would move it
B1	Predicted class I coverage is low and panel-dependent	instrument, partly disease	8.5% / 12.3% public junction, 27.4% / 30.4% pooled, on 10 / 34 alleles; 0.0% at a 0.37 cut; below a length-matched decoy null, per Section 7	**computational** — needs nobody's permission	Wider panels; a defended threshold; measured presentation
B2	Presentation predicted, never measured	instrument and access	Presentation percentiles on 174 peptides, from the predictor named in Section 2.2	**tissue + a proteomics facility** — the binding constraint on this route	Immunopeptidomics on EMC tissue or a patient-derived line
B3	Self-adjacency and central tolerance	disease	Lead peptide is 1 substitution (position 1, not an anchor under the general class I convention) from DMPCVQAQY in an *NR4A3* isoform, against a chance expectation of 0.02; 0 of 11 binders has an anchor-only near-self neighbour under that convention, 6 of 11 if position 1 counts, which is not established here	**a T-cell assay on matched donors** — needs material and a laboratory	T-cell reactivity assay against the specific peptide-HLA complex
B4	One strong CD4 epitope, on one allele of 23	instrument	44 binders, 1 strong (SYGQQNMPCVQAQYS on DRB1*14:01, 66.1 nM) over 14 of 23 alleles; class II coverage 6.5%, combined CD8 and CD4 1.8%	**computational**, and now largely spent	A class II threshold; measured presentation
B5	Seam-proximal peptides of four junctions occur in an *NR4A3* isoform	instrument, one failure mode resolved	170 of 174 novel proteome-wide; one binder withdrawn	**computational** — needs nobody's permission	Sequence novelty answered; cross-reactivity is not, and the filter should become isoform-aware
B6	Immunologically cold microenvironment	disease, addressable in combination	Quieter than comparator sarcomas on both archival cohorts; two individual-patient reports point the other way	**spatial profiling on tissue**	A vaccine supplies antigen; a checkpoint inhibitor supplies release
B7	Physical exclusion by the myxoid matrix	disease	Inferred from histology; the matrix pathway is now read in EMC and does not carry the analogy	**spatial profiling on tissue**	Vascular normalisation; matrix-directed agents
B8	EMC immune profiling is thin, not absent	access	Two archival cohorts read here; one published 12-tumour series, its spatial arm 2 specimens	**a larger series with spatial resolution**	Per-tumour spatial profiling at a scale that estimates an infiltrated fraction; class I as protein
B9	Manufacturing economics at this incidence	access	Five enumerable in-frame junctions, not per-patient discovery	**not this programme's to move**	A platform holder; a master-protocol vehicle
B10	Trial design below the randomisation threshold	disease	A 24-patient histology cohort has been run across nine centres	**a trial design decision, not a measurement**	Adaptive or histology-cohort design with a defensible endpoint

3.1 The limit worth watching

Measuring presentation is the most consequential next step (B2). Direct mass-spectrometry evidence of a fusion-junction peptide at measurable abundance on a common allele would turn a prediction into an observation. Evidence in any fusion-driven tumour would address the general premise; EMC-specific evidence would still be required for this route. The present coverage calculation estimates eligibility for an epitope whose presentation is unestablished.

No date is offered for this or any other capability. An earlier version gave optimistic, expected and conservative years for four capabilities. Those forecasts were withdrawn because readers could not check their basis. Section 6 instead states what would and would not count as the needed evidence.

B1. Low predicted class I coverage, dependent on the screen

Proposition. At the conventional MHCflurry threshold of 0.5 on the ten- and 34-allele panels, pooled global predicted coverage is below a third. This is a statement about those screening settings and pooled frequencies. Other thresholds, predictors and populations give different values (§2.3).

Evidence. The ten-allele screen gives four strong peptide–allele calls over three predicted presenting alleles, with coverage of 27.4%. The 34-allele screen gives five strong calls over four predicted presenting alleles, with coverage of 30.4%. Both panels contain the same four distinct strongly predicted peptide sequences; a peptide can generate a call on more than one allele. Conversely, two peptides called on HLA-B*15:01 do not add a second covered allele. Section 2.3 identifies every step and explains why five broad-panel calls produce four coverage steps. Section 7 reports the decoy comparison, against which this screen falls below expectation.

A high-confidence tier, and it is empty. The natural response to an undefended cut is to report a conservative tier beside it, and the conventional conservative cut for class I presentation is 0.2. At 0.2 this screen returns no presenting allele and no coverage at all: the least permissive call it makes is 0.3736, so every figure in this paper lives strictly between the conventional cut and a cut half its size. That is reported rather than suppressed, and it is the sharpest available statement of what these predictions are worth: they are not a set of confident calls with a permissive tail, they are entirely a permissive tail.

What would clear or move it. Three things, in ascending cost. Extending the panels beyond 10 and 34 class I alleles to the full validated set is a computational task and would raise the denominator.

The acceptance threshold is a convention rather than a result — nothing here defends 0.5 against 0.4 or 0.6 — and settling what the cut should be for junction peptides would do more to this figure than any panel extension. Calibrating that cut against a benchmark of experimentally validated neoepitopes is the form that settling would take.

The experimentally validated fusion-junction epitopes in the literature are individual sequences across a few fusions: the HLA-A*24:02-restricted SYT-SSX junction peptide [17,18], the four *EWSR1*::*FLI1* breakpoint peptides of the Ewing sarcoma case report [20] and the fusion neoantigens of a head and neck series [21].

A handful of epitopes is not a set against which a threshold can be calibrated. Calibrating on point-mutation neoantigens instead would import an assumption about junction peptides that is the very thing in question.

Measured immunopeptidomics (B2) could promote peptides the predictor ranks weakly or remove ones it ranks strongly, in a direction not knowable in advance.

Residual. Some fraction of patients will have no presented junction peptide and no second antigen to substitute. That fraction is a real and permanent exclusion from this approach, and it should be stated in any protocol rather than discovered at screening.

B2. Predicted rather than measured presentation

Proposition. No junction peptide has been shown to be presented on the surface of an EMC cell.

Evidence. All binding figures in Section 2 come from the sequence-based predictor named there [2]. No EMC immunopeptidomic dataset is known to the author, and the repository contains none. The vaccination reports in Section 1 bear on this weakly, because they measure T-cell responses to synthetic peptides that were administered, which is not evidence that the same sequence is processed from the endogenous fusion protein and displayed on a tumour cell [17,18,20].

Evidence outside the vaccination literature. In a series of 34 paediatric acute leukaemias, T cells expanded from patient bone marrow were co-cultured with autologous leukaemic blasts and the responding population sorted on activation markers, so no peptide was administered and the reactivity arose against antigen the tumour itself had processed and displayed [23].

Three recovered T-cell receptors then separated fusion from wild-type sequence on patient-matched alleles in cells given the fusion transcript rather than pulsed peptide: *KMT2A*::*AFF1* on HLA-DPA1*02:01/DPB1*01:01, a second *KMT2A*::*AFF1* breakpoint on HLA-DQA1*03:03/DQB1*03:01, and *PICALM*::*MLLT10* on HLA-B*51:01 from a CD8 clone.

Two of the three also killed leukaemic blasts from their own patients in culture, so the presenting cell was the tumour and not only an engineered surrogate. A fusion breakpoint therefore reaches a T cell by the endogenous route in humans, which earlier versions of this section recorded as unshown for any fusion.

What that evidence does not establish here. Its authors attribute the finding partly to anatomy: leukaemia arises in the tissues where T cells already reside, and the proximity makes engagement with blasts frequent. An EMC nodule is not that tissue, and a haematological malignancy speaks to neither microenvironment limit B6 and B7 record.

The fusions differ, and *KMT2A* breakpoints cluster in a way *EWSR1*::*NR4A3* need not.

The fusion-reactive clones sat at 0.0062%, 0.001% and 0.001% of the repertoire, were found only at diagnosis or at relapse and at no later timepoint despite deep sequencing, and one of the five samples taken through to receptor testing had no reactivity to either the fusion neoantigen or the blasts.

Recognition is established rather than abundance: no eluate was run and no copy number per cell was obtained, so the measurement below is unaffected, as are this paper's coverage figures and its grading of this route.

What would clear it. Mass-spectrometry immunopeptidomics on EMC tumour tissue or on one of the patient-derived EMC cell lines that have been established and characterised [6,14]. A single positive identification of a junction-spanning peptide in an EMC eluate would convert the central premise of this route from predicted to observed.

What a negative would have to specify to mean anything, and this paper cannot supply it. The sentence "a negative result would be close to decisive" appeared in an earlier version of this section and was doing no work, because a null eluate bounds nothing until three quantities are fixed in advance.

The first is a stated limit of detection in peptide copies per cell, so that "not presented" is separated from "below the instrument".

The second is a stated number of independent specimens and their HLA types, since the lead peptide is presented on one allele and a specimen not carrying it cannot test the claim.

The third is a positive control peptide of known abundance eluted in the same run, without which a null is indistinguishable from a failed elution.

This paper names those three requirements and does not meet any of them — naming them is not the same as having them. Their absence is why B2 is graded as bounded by access rather than by the disease, and it is what makes this the highest-value single experiment in the ledger.

Cost and owner. Requires tissue and a proteomics facility. It is not computational and cannot be performed by this programme.

B3. Self-adjacency and central tolerance

Proposition. The junction peptide is mostly self sequence with one or two novel residues at the seam, so the T-cell repertoire capable of recognising it may have been deleted or anergised.

Evidence. At the *EWSR1* exon 7 junction the seam carries a single novel codon formed from one leftover *EWSR1* nucleotide and two retained acceptor 5' untranslated nucleotides, after which *NR4A3* methionine 1 follows as an internal residue. The remainder of each peptide is parental sequence. Whether one altered residue is sufficient to break tolerance is not answerable by binding prediction.

Near-self search. We tested two questions that earlier versions had left unresolved: how closely each candidate resembles a normal human peptide, and where the differences occur. Anchor residues help a peptide bind HLA; other residues may contact the T-cell receptor and influence recognition. We searched all 11 predicted binders against the reviewed human proteome, including isoforms, allowing one or two substitutions. For each peptide, 50 residue shuffles provided a comparison for the number of near matches expected by chance.

Eight of the 11 binders have at least one human peptide within two substitutions. Every such neighbour lies in *EWSR1*, an *NR4A3* isoform or the paralogue *NR4A1*. That pattern is expected for a junction assembled from two self proteins. It checks the search as well as describing its result.

Three findings matter. **The lead peptide is one substitution from self.** NMPCVQAQY differs at position 1 from DMPCVQAQY, which occurs in *NR4A3* isoform 3 — the same isoform Section B5 is about — and at positions 1 and 5 from EMPCIQAQY in an *NR4A1* isoform.

The chance expectation is stated as a model rather than as a number: for a query q, draw N = 50 uniform random permutations of its own residues, search each against the same proteome at the same tolerance, and let $\mu(q)$ be the mean neighbour count over those draws. This holds length and amino-acid composition fixed and destroys only the order, so μ is the neighbour count attributable to q being a string of those residues rather than to q being that particular sequence.

For NMPCVQAQY, μ = 0.02 against an observed 2. **No p-value is quoted and none should be:** 50 draws bound a rate near zero loosely, the draws are not independent of the proteome's own composition, and the comparison this paper needs is an order of magnitude, not a tail probability.

Neither difference is at an anchor, under the convention applied here. Under this convention, position 1 and position 5 face outward or into the groove's middle; the primary anchors are position 2 and the C-terminus. The neoepitope and self peptide therefore differ at positions the T-cell receptor contacts. This is the less adverse configuration under the stated assumptions. A difference the receptor can read leaves open that a repertoire exists. A difference it cannot read does not. This distinction is the purpose of the filter.

An earlier version of this section stated the ordering the other way round; the withdrawal is recorded in Appendix C.

And no binder in the screen has an anchor-only neighbour under that convention. Zero of the 11 has a near-self peptide whose differences are confined to anchor positions, which would have been the worst case: an identical TCR-facing surface distinguished only by residues the T cell cannot see.

MPPPLRGDM's five neighbours are one *EWSR1* peptide, MPPPLRGGP, counted once per isoform, differing at positions 8 and 9 — and against a chance expectation of 1.34 for that peptide, five is not distinguishable from chance.

Three caveats carried with the result, and the first of them is load-bearing. The anchor assignment is the general class I rule of position 2 and the C-terminus applied uniformly, not an allele-specific motif, and the artifact marks every row so a reader can apply a different one.

Adding position 3, which HLA-A*01:01 reads as a primary anchor, to every peptide leaves the count at zero, so the caveat this paper had raised against itself does not bite.

Position 1 does. **Counting position 1 as an anchor — with or without position 3 — makes six of the 13 scored near-self hits anchor-only, across six of the 11 binders**, every one of them against the same *NR4A3* isoform, Q92570-3, that Section B5 is about. Three of those six differ from their self neighbour at position 1 alone, and the lead peptide NMPCVQAQY against DMPCVQAQY is one of the three; the other three differ at positions 1 and 2, and position 2 is a primary anchor under every convention.

Whether position 1 is an anchor for HLA-A*01:01, A*30:02, B*07:02, B*15:01, B*35:01 or B*44:02 is not established here, no allele-specific binding motif being held in this work, so the zero above is a result under a stated convention rather than a property of these peptides.

Second, sequence distance is not receptor distance: peptides three substitutions apart can present near-identical surfaces and peptides one apart can present different ones.

Third, the ordering of the two configurations holds where the near-self peptide is itself presented on the same allele, which is what an anchor difference would put in doubt, and presentation of these self peptides is measured nowhere in this work.

This search excludes one more failure mode — a close self peptide nobody had looked for — and leaves the question of whether a repertoire exists exactly where it was.

What the clinical prior art bears on it. Junction peptides from two other fusions have raised T-cell responses in patients: peptide-specific cytotoxic T cells against the SYT-SSX junction in synovial sarcoma [17,18], and de novo CD4+ responses against all four *EWSR1*::*FLI1* breakpoint peptides in one patient with Ewing sarcoma [20]. A repertoire able to see a fusion seam therefore survives central tolerance at some junctions. Neither result concerns this junction, these alleles, or this junction's single-residue distance to DMPCVQAQY; the SYT-SSX responses were weak and were not accompanied by demonstrated benefit; and the Ewing sarcoma observation is one patient. The limit is narrowed by an existence proof elsewhere rather than answered here.

What would clear or narrow it. Only a T-cell reactivity assay against the specific peptide-HLA complex answers the question. The search above changes what such an assay would be testing: the comparator is no longer a hypothetical self peptide but a named one, DMPCVQAQY, at a known single-residue distance, in a protein the tumour also expresses.

Comparator. This limit is where EMC differs most sharply from the melanoma setting, in which selected neoantigens frequently arise from point mutations in a repertoire that has not been tolerised against them and in a tumour already under immune pressure.

B4. One strong CD4 epitope, on one allele of twenty-three

Proposition. An effective vaccine generally requires CD4 helper epitopes. The corrected junction supplies one predicted strong class II binder on a 23-allele panel, restricted to an allele carried by about one person in fifteen. That is a statement about the screen, and it is reported here as one.

Why the result changed. We first rebuilt the class II candidates on the transcript model, so the class I and class II screens used the same corrected seam. The initial class II screen covered three DRB1 alleles and found no strong binder. The preceding version identified a wider DR, DP and DQ panel as the most likely way to change that result. The wider panel was then run and did identify a strong predicted binder. The previous negative therefore depended on panel coverage.

Result. At the corrected *EWSR1* exon 7 junction, 15 candidate 15-mers were screened with MHCnuggets [12] against 23 class II alleles — 15 DRB1, DRB3*01:01, DRB4*01:01, DRB5*01:01, two DPB1 and three DQB1 — calling a peptide a binder below 1000 nM and strong below 100 nM.

Every declared allele returned a score; none was unscreenable, so no cell of this panel is an absent reading.

44 peptide-allele pairs bind, spread over 14 of the 23 alleles, and one is strong: SYGQQNMPCVQAQYS on DRB1*14:01 at 66.1 nM.

The three-allele subset the previous version reported reproduces exactly within the wider run — YSQQSSSYGQQNMPC at 261.6 nM and QYSQQSSSYGQQNMP at 438.9 nM, both on DRB1*07:01 — so the widening is additive and the earlier negative was a property of the panel, not of a different computation.

What the positive does and does not bound, and it is less than it looks. The 15 candidates are single-residue-offset windows over one seam sharing 14 of 15 residues, so the number of independent peptides tested is nearer one than fifteen, and the 44 binders are overwhelmingly the same short stretch of sequence seen on different alleles.

One strong call on 23 alleles is a per-allele rate of 1 in 23, and the allele it lands on is not a common one.

Class II prediction is substantially less accurate than class I prediction. We therefore do not give the same weight to calls of equal nominal strength from the two classes. Class II peptides are not length-restricted by a closed groove, their binding register is not fixed, and the predictors are trained on far less measured data.

The single strong call should be read as a weaker statement than any single class I call in this paper, and no class II prediction here has been calibrated against anything measured.

What the result does establish is narrow and worth having: the junction is not devoid of predicted helper epitope, which is what three alleles could not distinguish from a junction that is.

What would clear or move it. Only measured class II presentation settles the prediction; a wider panel now has much less headroom to move it than it did.

How the two arms should be ranked, which is a separate question. Earlier versions of this section carried the class II result as the lesser of the two, and the human evidence does not support that ranking. Of the three fusion-specific restrictions recovered from endogenously presented breakpoints in paediatric leukaemia, two were class II and one class I [23], and the only de novo T-cell response reported to a fusion-breakpoint vaccine in a patient was CD4+ rather than CD8+ [20].

Three observations, across three fusions and three groups, are a pattern to test and not a rule: each restriction is one patient's alleles, none is in a solid tumour, and none concerns this junction.

The inverse ranking is no better supported and is not asserted here; what the pattern removes is the ground for assuming the CD8 arm is the one that matters.

The two predictions remain ordered, for the reason given above, and that ordering is a property of the models rather than of the responses.

A construct can also supply help from a heterologous source rather than from the junction itself, which is standard practice in peptide vaccine design and would make this limit a design constraint rather than a blocking one.

Consequence for the construct, and for the combined figure. Two things follow directly, and both changed with the panel. The candidate construct regenerated at the corrected junction now carries the CD4 epitope as well as the two strong class I epitopes NMPCVQAQY and QQNMPCVQAQY on HLA-B*15:01, and its minimal synthetic long peptide is 15 residues carrying both arms, where the class I-only version was 11.

And the combined CD8 and CD4 coverage figure can be computed at all for the first time: class II coverage is 6.5% on DRB1*14:01 alone, and the fraction of patients carrying at least one presenting allele of each class is 1.8%, the product of the two arms' coverages:

$$\mathbf{C_{I \wedge II} = C_I \times C_{II} = 0.2737 \times 0.0649 = 0.0178}$$

The class I term is the ten-allele screen's 27.4%, not the 34-allele screen's 30.4% quoted two paragraphs above: the instrument builds its class I allele set from the junction screen rather than from the broad-panel scan.

On the 34-allele set the same product gives 2.0%.

The product assumes independence between the class I and class II loci, the same assumption the within-class formula makes and no weaker. HLA-A, HLA-B and DRB1 sit on chromosome 6 in linkage disequilibrium, so it is an approximation whose direction is not known without haplotype frequencies.

The magnitude is bounded even though the direction is not, and by the same argument §2.3 applies to the within-class formula. This figure is an intersection rather than a union, so the bounds invert: under any dependence structure whatever it lies in [0.0%, 6.5%], and the upper end is the useful half.

A construct that needs a presenting allele of each class can never reach a larger fraction of patients than its scarcer arm alone, which here is the class II arm at 6.5%. That ceiling holds without any haplotype data, and it is 3.6 times the product rather than an order of magnitude above it.

It is a carrier-frequency product and nothing more: it says what fraction of patients carry a presenting allele of each class, not what a class II response contributes.

Class II help for a CD8 response is cognate rather than additive, and no model of that is offered here.

That figure is smaller than either arm alone and is the eligibility fraction for a construct that needs both — and, like every coverage number in this paper, it is a point on the curves of Section 2.3 rather than an estimate with a tolerance.

Status. Positive, on one allele of 23, with the previous negative explained as a panel artifact rather than overturned by a different method.

B5. Four peptides occur in an *NR4A3* isoform

Proposition, as originally stated. The junction peptides had been tested for novelty against two proteins rather than against the proteome, so their absence from normal human proteins was not established.

Result. All 174 distinct junction peptides were searched by exact substring against the UniProt reviewed human proteome, isoform sequences included. 170 are absent from every reviewed human protein. All 4 strong binders survive, including the *EWSR1* exon 7 lead NMPCVQAQY. The limit is largely cleared, and the sequence-level novelty premise of this route holds for the great majority of the peptide set.

The four that do not, and which junctions they belong to. DMPCVQAQ, DMPCVQAQY, DMPCVQAQYS and DMPCVQAQYSP all occur in Q92570-3, an isoform of *NR4A3* itself. One of them, DMPCVQAQY, is a predicted binder on HLA-B*35:01 at 369.1 nM. Those four peptides are not tumour-exclusive, and DMPCVQAQY is withdrawn as a candidate.

The pattern is a mechanism, and it survives the transcript model. A seam defined by exon boundaries is defined by whichever transcript declares them, and every figure in Section 2 comes from one canonical transcript per gene. The junction was rebuilt across all 99 protein-coding *EWSR1* transcripts against all 4 of *NR4A3*, at each of the five in-frame donor exons: 1,980 pairs, of which 970 emit an in-frame seam. The seam residue is not stable across them, taking nine distinct values — aspartate in 502 pairs, glycine in 251, asparagine in 118, serine in 59, no seam residue at all in 31, and four other residues in the remaining 9.

The collision appears in exactly the 502 aspartate-seam pairs and in none of the other 468. It is therefore not a property of the canonical annotation, and not a single case study: it is a property of the aspartate seam, reproduced independently in 502 transcript pairs, and its absence is equally determined. That is the mechanism this section had been asserting from one pair.

The pattern is not random across the junction set. All four collided peptides belong to the same four junctions, *EWSR1* exons 9, 10, 12 and 13, which are the four whose seam codon is aspartate on the canonical transcript; the *EWSR1* exon 7 junction has an asparagine seam codon and none of its peptides collides.

So four of the five in-frame junctions share a seam whose most seam-proximal peptides reproduce a normal *NR4A3* isoform sequence, and the one junction that is clean is the commonly reported public one that carries the lead binder.

Within the affected junctions the collision is confined to peptides that begin at the seam residue, which are the peptides with the least donor content; those extending further into *EWSR1*, including the strong binder RGDMPCVQAQY, remain novel.

The consequence for design is specific rather than general: for the aspartate-seam junctions, a construct should not rely on the seam-proximal window.

The methodological finding. The upstream novelty filter compares each candidate against the canonical parent proteins only, so an isoform that carries the seam sequence passes it unseen. That is a defect in the filter rather than in these particular junctions, and it will recur for any breakpoint whose seam residue reconstructs an isoform boundary. The proteome search reports this condition explicitly rather than silently discarding the hits.

A specific check for isoform collisions. A seam-proximal peptide begins with the seam residue and continues into the acceptor protein's sequence. It reproduces a normal protein when an acceptor isoform already places that same residue immediately before that same sequence. An alternative first exon can produce this arrangement.

We call the set of such residues the acceptor's *collision alphabet*. It is a property of the acceptor's isoform sequences. The resulting check is:

a junction whose seam residue lies in the acceptor's collision alphabet has a seam-proximal window that is not tumour-exclusive.

The alphabet is computed once per acceptor from its isoform set and answers the question for every donor and every breakpoint without a further search, where the proteome scan it anticipates costs one pass over the proteome per peptide. Here the alphabet is a single residue, aspartate, contributed by Q92570-3.

It is exact on every transcript pair this locus can produce. Applied to the 970 in-frame pairs of the 1,980 above, the one-residue test agrees with the proteome search on all 970: 502 predicted collisions, all observed; 468 predicted clean, none colliding; no disagreement in either direction. That the two ends match is the point. A rule that flagged the aspartate seam would be unremarkable if it also flagged the glycine seam's 251 pairs or the asparagine seam's 118, and it flags neither.

And it does not replace the search, which is the failure mode from the other side. The test anticipates one condition: a seam-proximal peptide reconstructing an acceptor isoform boundary. It is silent about a peptide colliding with an unrelated protein, and silent about peptides extending far enough into the donor to become novel again — this locus's own strong binder RGDMPCVQAQY carries the same stem, begins three residues 5' of the seam, and survives. A filter that ran this instead of the proteome search would reintroduce the defect it was built to catch. It is a pre-screen that says where the search will find something, and it costs one comparison.

What a clean result does not license. A peptide absent from every reviewed human protein is not thereby safe. A T-cell receptor engages a peptide-MHC surface rather than a sequence, so a peptide differing from a self peptide at a position that does not contact the receptor can still be cross-recognised.

Unreviewed sequences were searched separately, over 127,090 entries: of the 170 peptides absent from every reviewed protein, 12 occur in at least one unreviewed entry, and none of those 12 is a predicted binder.

A hit among predicted-and-unreviewed entries is not evidence that a normal protein carries the peptide, and a miss there is not evidence of absence, so this withdraws no peptide and confirms none; what it establishes is where the sequence-novelty premise is weakest, and that is not where the candidates are. This test excludes one specific failure mode and leaves the others standing.

B6. Immunologically cold microenvironment

Proposition. EMC has a low mutational burden and a sparse infiltrate, so there is little pre-existing antigen-specific response for an intervention to amplify. The mutational-burden half is not disputed here. The infiltrate half was an assumption when this section was first written and is now a measurement, reported at B8 and not restated here.

What that measurement does to this proposition, and what it does not. It supports the direction B6 asserts and does not license the word cold, because the comparator is other sarcomas rather than an immune-normal reference, and because a bulk average over whole tissue cannot tell a tumour with no T cells from one whose T cells are held outside it. That second reading is B7's, the two are not the same claim, and B7 states why the evidence now available does not choose between them.

Evidence, and an observation about this programme's own grading. This proposition is why this programme set most immune-modulating classes aside for this disease in its route ledger, a committed record of forty candidate modalities each carrying a verdict and a reason. Those reasons are quoted below from that ledger: they are this author's own prior notes, not positions taken in the literature.

Innate agonists of the STING, TLR and RIG-I classes were excluded there because such agonism "supplies the danger signal and the priming context; it does not supply antigens, and a genome as quiet as EMC's is short of antigens rather than short of priming".

In-situ vaccination was excluded because it "releases and adjuvants the antigens the tumour already has", so "releasing more of nothing does not help".

Checkpoint inhibitors beyond PD-1, costimulatory agonists, adenosine-axis inhibitors and regulatory T-cell depletion were each excluded for acting on a response presumed absent.

Every one of those is an argument about antigen supply, and a vaccine is an antigen supply, so none of them argues against the class in a configuration where antigen is supplied exogenously. That is why the combination in Section 4 was never graded here as a unit.

The symmetry is not exact, and an earlier version of this paper asserted that it was.

The vaccine was not parked for want of priming: its ledger entry is on the board rather than excluded, and its stated reason for being parked is immunogenicity, "which its own record states is not a computational question".

The standing blocker against it names two things, that EMC is antigen-cold and that the fusion junction is a weak peptide-HLA, and the second is an antigen-side objection no priming-directed partner answers.

The correct statement is the weaker one: a partner supplies what several exclusions called missing, and does not supply what the vaccine's own blocker calls missing.

What would clear it. Clinical evaluation of a vaccine together with an agent that supplies priming or releases inhibition. Section 4 sets out the specific backbone for which EMC evidence already exists.

B7. Physical exclusion by the myxoid matrix

Proposition. EMC is immune-excluded by a dense chondroitin-sulfate gel, which is a physical barrier rather than a signalling programme.

Evidence. The myxoid matrix is the disease's defining histological compartment. The oncofetal chondroitin sulfate pathway is the mechanism proposed for it here by analogy: the glycosaminoglycan biosynthesis genes of that pathway are differentially expressed and correlated with immune response in placenta and colorectal cancer [8], which is the tissue setting that study examined.

That pathway has since been read in the same two EMC cohorts and the reading does not carry the analogy across. The linker and core-protein modules are higher in EMC than in the comparator sarcomas on both platforms, the backbone-polymerisation and 4-O-sulfotransferase modules disagree between platforms, the sulfate-donor module is lower on both, and three sulfotransferase modules fall below the read's own readability floor.

Values have one home, `research/modalities/emc-expression-panels.json`, read 2.

A transcript level is a capacity to build a glycan and not the glycan, so the inference from those tissues to this one remains the author's.

Transforming growth factor beta inhibition, which is the standard proposal for immune-excluded tumours, was set aside for EMC precisely on the ground that the exclusion here is physical rather than driven by a fibroblast programme.

Why this differs from B6. A cold tumour lacks an effective immune response. An immune-excluded tumour may contain a response that cannot reach the tumour cells. These require different interventions. A vaccine could increase circulating junction-specific T cells without enabling them to enter the tumour.

Why the new EMC immune evidence does not choose between them. Both instruments that speak to the question arrived in 2026 and neither sees what the other sees.

The bulk cohort reading of B8 runs in the direction B6 predicts, and it is an average over whole tissue with no spatial resolution, so a cold tumour and an excluded one return the same value.

The multiplex immunofluorescence arm of the twelve-tumour series runs in the direction B7 predicts, because a T cell that is present and suppressed is not a T cell that is absent, and it is two specimens compared within themselves and called proof of concept by its own authors [15].

The instrument with a cohort has no spatial resolution and the instrument with spatial resolution has no cohort.

This paper therefore does not resolve the disagreement and does not treat either limit as retired by the other's evidence; what would resolve it is the observation B8 names, per-tumour spatial profiling on a series large enough to estimate a fraction.

What would clear or mitigate it. Vascular normalisation is the mechanism with the most direct EMC evidence, discussed in Section 4. Matrix-directed approaches, including addressing the oncofetal chondroitin sulfate modification itself, are registered in this programme as candidates and are not resolved.

B8. Thin EMC immune-profiling data

Proposition. The characterisations of EMC as cold and as excluded were inferred from the disease's mutational burden, its histology and sarcoma-wide immunotherapy experience rather than from EMC-specific immune profiling. The proposition as originally stated, that no such profiling exists, no longer holds; what follows replaces it.

Consequence. Several exclusions above rested on an assumption nobody had measured in this disease, and an absent reading was never a reading of absence.

What has since been read, in EMC tumour tissue. Two archival bulk expression cohorts, 6 and 10 EMC tumours against 29 and 6 comparator sarcomas, were scored against lineage, effector and antigen-presentation modules and against published hallmark infiltration proxies.

The interferon-γ effector axis, the myeloid compartment and class II antigen presentation are lower in EMC than in the comparator sarcomas on both platforms; the cytotoxic-lymphocyte and T-cell modules are lower on the larger cohort; and two hallmark proxies, the interferon-γ response and the lymphocyte-weighted allograft-rejection set, are lower on both.

The class I apparatus is the exception. β2-microglobulin and the class I heavy chains sit near the top of the expressed range on the larger platform, and HLA-A, HLA-C and TAP1 read flat against the comparator rather than lost.

Values and their contrasts have one home, `research/modalities/emc-expression-panels.json`, reads 8 and 19.

What points the other way, in individual patients. Twelve archival EMC tumours profiled by whole-transcriptome targeted sequencing carry more B-cell infiltration in the low-risk half of the series than in the high-risk half.

Proof-of-concept multiplex immunofluorescence on two of the specimens reports exhausted CD3+CD8+PD1+ T cells and FOXP3+ regulatory T cells in the high-risk one.

The authors state that every immunofluorescence comparison is a within-specimen region-of-interest analysis, that their prognostic model overfits, and that the work is exploratory and hypothesis-generating [15].

A single reported case profiled as immune-enriched despite a low mutational burden is graded in Section 4 [16].

Neither is a cohort estimate, and a quiet distribution with occasional infiltrated members is consistent with both those observations and with the paragraph above.

What the twelve-tumour series adds beyond direction is that the variation it reports is within EMC rather than between EMC and something else, which is the axis a patient-selection argument would have to run on.

What none of it settles. The comparator arm is other sarcomas, which are not an immune-normal reference, so lower than that comparator is not the same statement as cold.

Bulk expression cannot locate a transcript in a cell, and therefore cannot separate a cold tumour from the excluded one B7 describes.

Sixteen tumours cannot estimate what fraction of EMC is infiltrated, which is the quantity a patient-selection argument would need.

The class I reading is transcript abundance in a cohort, not protein on a cell surface in a patient, and class I loss would independently disable every antigen-directed route including this one.

And the twelve-tumour series has been read here only as its published abstract, because the publisher returned an access error to this work's retrieval runner, so its infiltration method, its external validation cohorts and its per-specimen values are unexamined.

What would still clear it. Per-tumour profiling with spatial resolution on a series large enough to estimate that fraction, and class I assessed as protein.

Cost and owner. The cohort reading is done. It ran on a continuous-integration runner over two public archival series, with no rental, no accelerator and no purchased compute. What remains requires tissue, and it is still the cheapest tissue-based item in the ledger and the one that most efficiently informs the others.

B9. Manufacturing economics at this incidence

Proposition. An individualised vaccine requires per-patient sequencing, design and manufacture, and EMC's incidence is well under one per million per year.

Consequence. No independent programme can supply the manufacturing, and the economics that support an individualised product in melanoma do not obviously extend to a disease with this incidence.

What would mitigate it. Two features of this target reduce the requirement relative to a general individualised product. First, the antigen is not discovered per patient by tumour sequencing but is determined by which exon pair the patient carries, of which five are in frame; the design space is therefore small, enumerable in advance, and shared across patients with the same breakpoint.

Second, because the fusion is truncal, the antigen does not need re-selection over the disease course.

A small fixed panel of breakpoint-specific constructs, allocated by a diagnostic assay, is closer to a stratified product than to a bespoke one.

A construct of that shape has been made and administered once already: the Ewing sarcoma vaccine was an off-the-shelf multi-peptide product spanning a recurrent breakpoint rather than a per-patient design [20].

Routine fusion analysis in a clinical genomics workflow finds EWS breakpoints clustered tightly, with about three quarters of Ewing sarcoma and desmoplastic small round cell tumours sharing one exon 7 motif [22].

Whether that is commercially tractable is a question for a platform holder and not one this analysis can answer.

B10. Trial design below the randomisation threshold

Proposition. At this incidence a conventional randomised trial of a vaccine addition is not feasible.

Evidence and mitigation. A histology-specific EMC cohort within a sarcoma master protocol has already been executed and reported for a different regimen, enrolling 24 patients across nine centres in three countries over four years [5]. That shows the vehicle exists, and it gives an accrual rate of about six unselected patients a year across nine centres.

And that rate does not survive this paper's own eligibility filter. A junction-vaccine cohort cannot enrol unselected EMC: it needs *EWSR1*::*NR4A3*, which is 62 to 75% of cases [7], and then an HLA type that presents a junction peptide, which is 8.5 to 30.4% of those (B1).

Applying both fractions to that trial's own accrual leaves roughly 0.3 to 1.4 eligible patients a year, so a cohort of the size already run would take on the order of two decades, and a powered comparison considerably longer.

That is what makes B10 a limit of the disease rather than of study design: the vehicle exists and the patients to put in it do not arrive fast enough.

The endpoint question is separate — response-based endpoints behave poorly in indolent tumours, and the existing EMC cohort reported progression-free survival at a fixed time point.

4. An ungraded combination

The vaccine and a potential partner would address different limitations. A vaccine cannot by itself resolve the microenvironment barriers in B6 and B7. A checkpoint inhibitor cannot by itself resolve the antigen-related barriers in B1 and B3. This motivates a narrower research question: could a junction vaccine add anything to a treatment backbone with EMC-specific activity? The prior negative assessment of the vaccine alone does not answer that combination question.

Such a backbone exists. A phase 2 histology-specific EMC cohort within the IMMUNOSARC II master protocol evaluated sunitinib with nivolumab in adults with advanced, progressing, centrally confirmed EMC across nine centres in Spain, Italy and the United Kingdom.

Of 23 evaluable patients, 16 were progression-free at 6 months, median progression-free survival was 13.2 months (95% CI 5.7 to 20.7), and there were 2 partial responses [5].

This is a conference abstract and has not been peer-reviewed; the full publication is not available, and no results are posted for the registration.

It is reported here as the most direct EMC-specific evidence available for an immunotherapy-containing regimen, and not as evidence of efficacy. The preceding mixed-histology phase Ib/II study of the same combination is published [4].

Three features make this backbone the natural context for the vaccine question.

The antiangiogenic component addresses B7 by a mechanism: vascular normalisation reduces the physical and vascular barriers to lymphocyte entry.

Antiangiogenic tyrosine kinase inhibitors carry the most consistent prospective signal in this disease — pazopanib gave an objective response in 4 of 22 evaluable patients with a median progression-free survival of about 19 months [11], and a sunitinib series reported activity in translocated EMC [13].

The checkpoint component addresses the release arm of B6. Neither addresses antigen supply, which is what the vaccine would contribute.

One reported single-agent checkpoint response, and what it changes. The checkpoint half of that backbone has never been given alone in a reported EMC cohort, and one patient has been reported outside one. The patient was a man with *EWSR1*::*NR4A3* EMC, microsatellite stable at a tumour mutational burden of 0.67 mut/Mb and with *TSC2* loss, whose microenvironment a commercial assay profiled as immune-enriched and fibrotic with high PD-L1.

He received single-agent pembrolizumab with a near-complete response reported over 15 cycles [16].

One patient in an industry-authored conference abstract that has not been peer reviewed supports no response rate, and it is not read here as a promotion of the checkpoint arm.

What it does is name the selection axis: the markers conventionally used to choose a patient for checkpoint inhibition and the markers describing that patient's microenvironment pointed in opposite directions in the same tumour, which is the heterogeneity B8's cohort reading permits and cannot measure.

That is a question for B8's observation to settle, not a case for adding a third agent.

This is also the architecture of the melanoma programme behind the recent announcement, where the individualised vaccine is given with pembrolizumab and never alone [3,9]. The transfer to EMC fails on antigen depth — melanoma supplies a pool of private neoantigens, EMC one junction — and not on architecture.

What this section is claiming, and what it is not. The proposed combination joins a breakpoint-matched junction construct to a checkpoint and antiangiogenic backbone. The rationale is that each component addresses a limitation the others do not. The combination has not been graded as a unit, including in this programme's own ledger, which is the source of B6's observation.

It is not a trial proposal.

No population, comparator, endpoint, effect size or sample size is specified here; a single-arm addition to a backbone whose own six-month progression-free rate is 16 of 23 could not attribute an effect to the vaccine;

and B10's arithmetic says the eligible patients do not arrive at a workable rate.

Whether the combination merits evaluation at all turns on B2.

Nothing here recommends that any patient receive any of these agents outside a clinical trial.

5. Present work and pending arrivals

Four items stood on this list in the previous version, all needing nothing but public data. Two have since been completed. The distance-to-self and anchor-versus-contact-position filters form the near-self search in B3. Extending the class II panel from three DR alleles to 23 across DR, DP and DQ produced the positive prediction in B4. These are reported results, not promised future work.

The class I panels remain at 10 and 34 alleles against the full validated set, which would raise the denominator of every coverage figure.

A defended acceptance threshold for junction peptides is still absent, and Section 2.3 now shows exactly how much rests on it.

The isoform-aware novelty filter identified in B5 is still unbuilt. None of the remainder requires permission, material or funding.

Everything else waits. Measured presentation (B2) would change the character of this route rather than its score, because the screen in Section 2 would stop being a stand-in and become a hypothesis with a calibrating dataset.

B8 is no longer an absence: two archival EMC expression series have been read here and a twelve-tumour series has been published, and what remains of that limit is spatial resolution on a series large enough to estimate an infiltrated fraction, together with class I assessed as protein rather than as transcript.

It is still the cheapest tissue-gated item and the one that informs the most others, because infiltrate location and HLA class I status bear on every antigen-directed route in this disease.

Access to a patient-derived line would additionally make B2 executable rather than merely specified; these are separate arrivals and neither implies the other.

B6 and B7 are properties of the tumour that no computational advance mitigates.

6. Distinguishing a real advance from an apparent one

A new result should be judged against the particular gap it is meant to close. The examples below distinguish evidence that would change this assessment from evidence that would leave the central question unresolved.

A result on a different fusion is not a result on this one. Immunogenicity studies on other fusion-driven tumours already exist and raise the prior that fusion junctions in general can be seen by T cells [17,18,20,21], and one of them shows a breakpoint reaching a T cell after processing from the endogenous transcript rather than from an injected peptide [23]. None establishes that this junction, on these alleles, is presented at usable abundance, and none of them measures abundance at all. The discriminating question to ask of any such report is whether its evidence concerns abundance, processing, or only a different fusion; the best of the reports available today reaches processing, and step 3 of Section 6.1 turns on abundance.

Robotic execution is not material access. A cloud laboratory with per-experiment pricing supplies robots and generic reagents, not an EMC line or organoid, and without that none of the tissue-gated items in Section 3 becomes runnable.

A better predictor is not a presentation measurement. An improved class I or class II model could change the numbers in Section 2. B2 would remain unresolved because no EMC presentation measurement has been made.

A larger allele panel raises a figure without grounding it. Extending the panels can only move coverage upward, and a higher predicted-presentation figure is not evidence that any patient's tumour presents anything.

6.1 Order of the remaining questions

The remaining questions can be addressed in an order that avoids unnecessary work. Each step is less costly than the next and can close the route. No dates are proposed, and none of these steps is scheduled.

1. **Defend the acceptance threshold, or record that it cannot be defended.** Computational, needs nothing but public data. Calibrate the cut against experimentally validated epitopes, and if no validated set restricted to fusion-junction peptides exists, that absence is itself the finding and Section 2.3's curve is the only honest report of coverage. Until this is done every figure in B1 is a point on a curve.
2. **Measure whether the lead peptides bind their alleles at all.** Four peptide-allele pairs carry every class I figure in this paper; Table 3 ranks them by what each costs if it fails. Synthesise them and test binding in a cell-based stabilisation assay, first pair first.

Table 3. The four peptide-allele pairs every class I figure rests on, ranked by what each costs if it fails. *Coverage at risk* is a leave-one-out difference — the pooled figure with all four presenting alleles minus the figure without this one — and not the allele's marginal contribution as it enters Section 2.3's curve, which is order-dependent and larger. *Percentile* is the call that put the allele in the set. Testing binding for these four is the only step in Section 6.1 that needs neither an EMC specimen nor a proteomics facility.

Order	Peptide	Allele	Percentile	Coverage at risk
1	RGDMPCVQAQY	HLA-A*01:01	0.4061	9.9 pp
2	MPPPLRGDM	HLA-B*07:02	0.458	7.2 pp
3	NMPCVQAQY	HLA-B*15:01	0.3736	6.5 pp
4	NMPCVQAQY	HLA-A*30:02	0.4033	3.0 pp

3. **Look for the peptide on a tumour.** Immunopeptidomics on EMC tissue or a patient-derived line, with the limit of detection, the specimen count and their HLA types, and the abundance-matched positive control all fixed before the run, per B2. This is the decisive step and the first that requires material this programme cannot obtain.
4. **Only if step 3 is positive, ask whether a T cell sees it.** Recognition of the peptide-HLA complex by T cells from HLA-matched donors. A negative here is the one that B3 predicts and that nothing computational can rule out.
5. **Only then is there anything to build.** The construct in Section 2 is an output of a screen and is not a candidate for administration; it exists so that steps 2 to 4 have something specific to test.

Steps 1 and 2 would together decide whether the rest is worth commissioning, and neither needs an EMC specimen. That is the whole of what this paper can say about what to do next.

6.2 Conditions for revision

Four statements in this paper are falsifiable by an observation a reader could go and make, and they are listed so a future reader can check them rather than re-derive them.

Sequence-level novelty for 170 of 174 peptides would be overturned by a proteome release adding a protein that contains one of the four strong binders.

Every coverage figure would move upward if a wider panel added a presenting allele and downward if measured presentation removed one of the strong calls, and all of them go to zero if the acceptance threshold is set below 0.37.

The class II statement said, in the version this paper supersedes, that it would be changed by a wider DR, DP and DQ panel — named there as the single most likely source of a change to it.

That panel has since been run and it did change it, from no strong epitope on three alleles to one on 23; the statement in B4 is now the wider panel's, and what would change it again is measured class II presentation rather than more prediction.

And the argument in Section 4 would be closed altogether by an EMC series showing HLA class I loss as protein; the transcript-level reading in B8 does not show it, and a protein-level series that did would end every antigen-directed route in this disease including this one.

7. Limitations

All binding figures are predictions from sequence-based models and no EMC-specific validation of either model exists. They predict peptide-MHC affinity alone: proteasomal cleavage and TAP transport are not modelled here, so a strong call bounds what could be presented rather than naming a peptide that is.

The headline coverage calculation multiplies non-carrier frequencies across alleles. It assumes independence between loci and approximates the dependence between alleles at the same locus. Two pooled alleles are both HLA-B; accounting for their shared locus moves the pooled value by about 0.3 percentage points. Actual dependence between loci was not measured or estimated from haplotype data. Possible coverage can nevertheless be bounded using the available marginal frequencies: §2.3 gives [17.5%, 29.9%] while retaining the within-locus assumption, and [9.2%, 31.1%] after also relaxing that assumption. Haplotype or genotype information could estimate the actual dependence or narrow the corresponding bounds. These bounds are not confidence intervals and do not quantify uncertainty in the peptide predictions.

The population-to-region mapping is an approximation, regional figures are point values on samples as small as 579 individuals, and the frequencies are pooled across populations whose allele frequencies differ by more than the figures being reported.

A fourth dependence sits underneath all of these and was not measured until now: every peptide in Section 2 is derived from one canonical transcript per gene, and across the 970 in-frame transcript pairs tested under B5 no junction peptide is common to all of them.

The seam residue itself takes nine values, so the peptide identities move with it.

The counts are more stable than the identities — each pair yields 30, 34 or 38 novel peptides and no other value — but a reader should take the specific sequences named here, NMPCVQAQY included, as conditioned on the canonical annotation rather than as properties of the locus.

Which transcript a patient's fusion actually uses is not decidable from annotation and is not decided here.

The binder counts and every coverage figure derived from them depend on an acceptance threshold this paper does not defend, and no multiplicity correction is applied anywhere.

The screen tested 174 peptides against 10 and then 34 alleles at two thresholds.

It now has a decoy null, and it sits below that null.

Ten length-matched random peptides were drawn per junction peptide from the same reviewed human proteome the novelty search uses — 1,740 in all, matching the screen's own 36 8-mers, 41 9-mers, 46 10-mers and 51 11-mers — and were scored on the 34-allele panel through the same presentation percentile.

They pass at the conventional cut on 0.644% of 59,160 peptide-allele tests, exact 95% confidence interval 0.581% to 0.712%, which is also a reading of the instrument: a percentile of 0.5 admits more than the 0.5% its name implies, and the interval excludes 0.5%.

Resampling 174 of those decoys 2,000 times, a random set of the screen's own size presents on a median of 23 of the 34 alleles, a mean of 22.6 and a minimum of 7, and the closed form for that pass rate gives 22.95, so the figure is arithmetic rather than resampling noise.

This screen presents on 4 of them, and no random set fell that low.

The coverage figures are therefore not an enrichment over chance in either direction: the deviation is negative, and these junction peptides are a worse source of predicted class I binders than arbitrary human sequence of the same lengths.

That bounds the screen and not any peptide in it.

A set can fall below a random background and still contain real binders, so the null does not say the four strong calls are false, and predicted binding against predicted binding is not evidence about presentation, immunogenicity, safety or benefit.

Three things bound the null itself.

It is computed at the same undefended cut as the calls it is a null for, so it inherits that limitation rather than removing it.

Its decoys are arbitrary peptides of matched length rather than composition-matched shuffles of the junction peptides, so it does not separate this junction from the amino-acid composition of the peptides spanning it.

And its decoys are drawn from the proteome those peptides were filtered against for novelty, so they are self peptides — the background the percentile scale is itself defined against, and not a background of neoantigens.

The near-self search of B3 carries its own null for the same reason a binding screen needs one: a count of near-self neighbours is meaningless without the number a peptide of that length and composition finds by chance.

The IMMUNOSARC II EMC cohort result is a single-arm conference abstract that has not been peer reviewed. Of the combination's components, the tyrosine kinase inhibitor has the larger independent EMC evidence base. We cite the result to establish that a trial vehicle and a backbone exist, not to attribute activity to the immune arm.

The single-agent checkpoint response cited beside it is one patient in an industry-authored conference abstract and bounds nothing.

EMC's characterisation as immune-excluded rests on inference; its characterisation as immunologically quiet rests on the two archival cohorts read here against comparator sarcomas, which is a contrast and not an absolute, and on one twelve-tumour series read here only as an abstract.

Neither instrument separates a cold tumour from an excluded one, which is limit B8 and the reason B6 and B7 remain graded apart.

The clinical prior art for fusion-junction peptide vaccination is two small early-phase synovial sarcoma trials whose immunological readouts were weak and one single-patient Ewing sarcoma report [17,18,19,20]; it bears on whether such a construct can be administered and immunologically monitored, and on nothing about benefit in any disease.

The class II panel is 23 alleles across DR, DP and DQ, its one strong call lands on a single uncommon allele, and class II prediction is less accurate than class I, so B4's positive bounds less than a class I call of the same nominal strength.

No claim is made that any peptide is presented, that any construct would be immunogenic, that any combination would be safe or effective, or that any of this is ready for clinical use.

No wet-laboratory work was performed, and the measurements this characterisation most needs require work this programme cannot carry out.

Figure legends

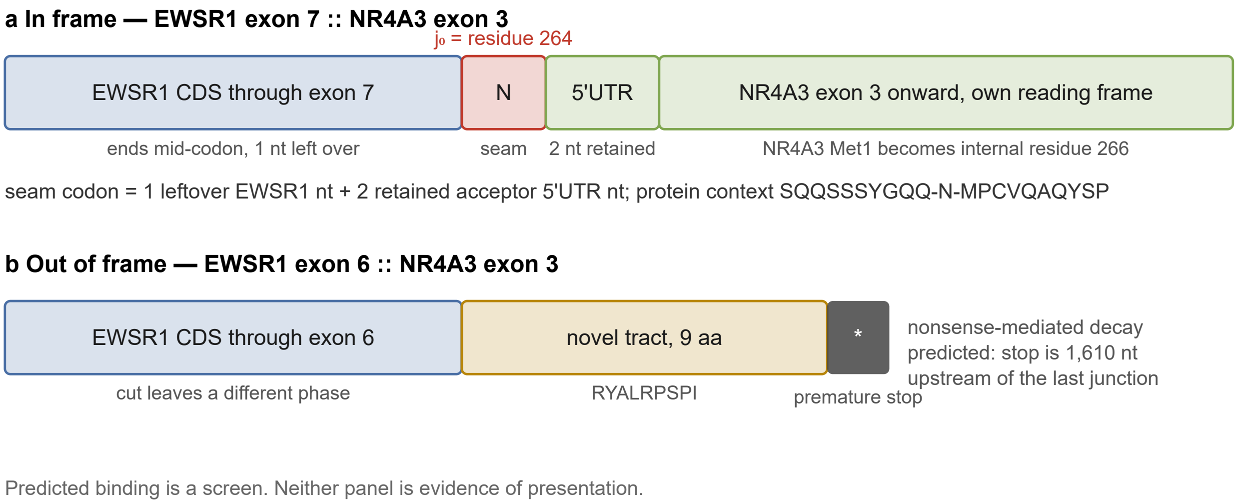

Figure 1. The seam codon is built from leftover donor nucleotides and the acceptor exon's retained 5' untranslated region, and its identity decides whether the isoform collision occurs. (a) The *EWSR1* exon 7 to *NR4A3* exon 3 junction. Donor coding sequence ends mid-codon with one nucleotide left over. Two retained acceptor 5' untranslated nucleotides complete the codon, giving the seam residue at $j_0 = 264$. *NR4A3* then resumes in its own frame, with its methionine 1 as internal residue 266. Four of the five in-frame junctions place aspartate at this position rather than asparagine, and that is the difference Section B5 is about. (b) An out-of-frame junction reads the same acceptor exon in a shifted register: 9 novel residues, then a premature stop 1,610 nucleotides upstream of the chimera's last exon-exon junction, which is the canonical nonsense-mediated-decay configuration. Predicted binding is a screen; neither panel is evidence of presentation.

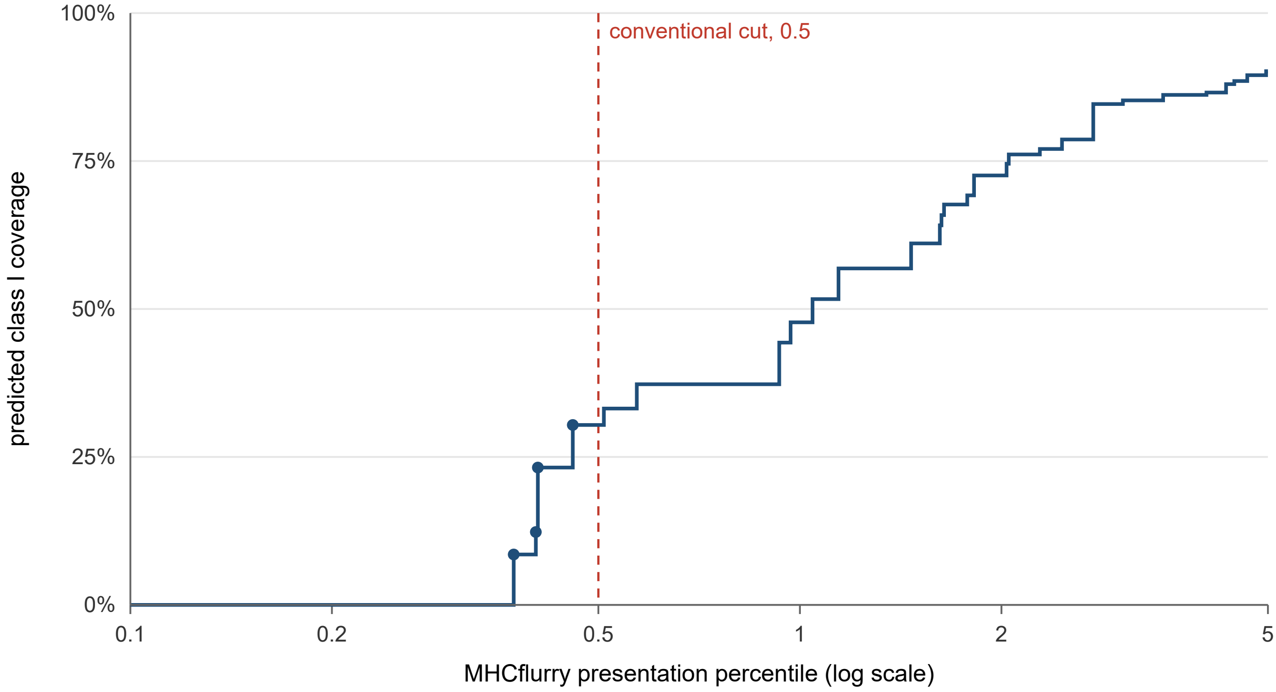

Figure 2. Predicted class I coverage against the acceptance threshold, log axis. The step function over the 34-allele panel, each step one peptide-allele call, drawn to the largest threshold the predictions can speak to. Filled circles mark the four steps below the conventional cut, which span 0.0844 percentile units; the dashed line is that cut, drawn as one annotated vertical rather than as the plot's endpoint. The axis is logarithmic because the function's shape occupies two decades and a linear axis collapses every step the argument rests on into the left margin. Coverage is the union carrier frequency of the presenting alleles on pooled AFND frequencies.

8. Methods and reproducibility

Every figure in Sections 2 and 3 is generated by a script in `research/modalities/` and is committed as a JSON artifact beside it. The junction and coverage figures come from `fusion_breakpoints.py` (the junction set and predicted binders), `hla_coverage.py` (population coverage), `coverage_scan.py` (the coverage-versus-allele-count curve) and `coverage_threshold_curve.py` (the coverage-versus-threshold curve of Section 2.3).

The screens come from `junction_proteome_novelty.py` (the proteome search of Section B5), `junction_frameshift_peptides.py` (the out-of-frame screen of Section 2.2), `junction_selfsimilarity.py` (the near-self search of Section B3) and `predictor_concordance.py` (the second-predictor check).

The rest come from `coverage_uncertainty.py` (the within-locus form, the Fréchet bounds and the between-population spread of Section 2.3), `novelty_seam_test.py` (the one-residue pre-screen of Section B5 and its validation), `patient_neoepitopes.py` and `patient_cd4_epitopes.py` (the per-patient shortlisters) and `vaccine_construct.py` (the candidate construct and its minimal synthetic long peptide).

The corresponding artifacts are `fusion-breakpoint-neoantigens.json`, `hla-coverage.json`, `coverage-curve.json`, `coverage-threshold-curve.json`, `epitope-allele-matrix.json`, `epitope-allele-loose-matrix.json`, `junction-proteome-novelty.json`, `junction-frameshift-peptides.json`, `junction-selfsimilarity.json`, `predictor-concordance.json`, `coverage-uncertainty.json`, `novelty-seam-test.json`, `patient-cd4-demo.json` and `vaccine-construct.json`.

The two tables in this paper are generated from those artifacts by `vaccine_path_tables.py` and a build fails if a cell and its source diverge.

The predictor versions those artifacts record are MHCflurry 2.1.4 with models release 2.2.0 for class I and MHCnuggets 2.4.1 for class II.

MHCnuggets publishes no models-release identifier separate from its package version — it ships its trained weights inside the distribution — so the version is the whole of the provenance available for that arm, and the artifact records why rather than leaving the field blank.

Table 2. The 34-allele class I panel. Listed so the coverage scan can be reproduced and the choice of panel assessed rather than taken on trust. It is a common-allele HLA-A and HLA-B panel spanning global diversity, and it carries no HLA-C and no class II allele — so every figure computed on it is a partial coverage of the class I repertoire, and §2.3's statement that extending the panel can only raise the figures applies to HLA-C before it applies to anything else.

Table 2

Locus	Alleles
HLA-A (16)	HLA-A*01:01, HLA-A*02:01, HLA-A*02:03, HLA-A*02:06, HLA-A*03:01, HLA-A*11:01, HLA-A*23:01, HLA-A*24:02, HLA-A*26:01, HLA-A*30:01, HLA-A*30:02, HLA-A*31:01, HLA-A*32:01, HLA-A*33:01, HLA-A*68:01, HLA-A*68:02
HLA-B (18)	HLA-B*07:02, HLA-B*08:01, HLA-B*13:02, HLA-B*14:02, HLA-B*15:01, HLA-B*18:01, HLA-B*27:05, HLA-B*35:01, HLA-B*38:01, HLA-B*40:01, HLA-B*40:02, HLA-B*44:02, HLA-B*44:03, HLA-B*46:01, HLA-B*51:01, HLA-B*53:01, HLA-B*57:01, HLA-B*58:01

These are not regenerated on every commit, and this paper does not claim they are. The workflow that runs them is dispatched by hand, its steps do not fail the run when a generator fails, and it writes its outputs to a separate cache branch from which they are copied in. So the artifacts in the repository are the record of a run that happened, verifiable by their embedded timestamps and input hashes, rather than the output of continuous re-execution. An earlier version of this section said they were regenerated in continuous integration; that was not true of any branch a reader would fetch.

The decoy null of Section 7 is generated by `vaccine_threshold_calibration.py`, whose artifact is `vaccine-threshold-calibration.json`; it imports its peptide lengths and its acceptance cut from `coverage_threshold_curve.py` and its proteome fetch from `junction_proteome_novelty.py`, so the null and the screen it is a null for cannot be computed against different conventions.

Clinical figures are quoted from the curated EMC registry at `research/data/emc-clinical-registry.json`, which carries a structured citation entry for every source and a retrieval note for those whose identification required one.

9. Declarations

Use of AI tools. A large language model (Claude, Anthropic) was used throughout this work: to write the analysis code, to run the screening pipelines, to draft and revise this manuscript, and to conduct the internal adversarial review of earlier drafts whose findings this version incorporates. The model versions used over the span of the work are the ones the repository's commit record names, and are not restated here. No quantitative result was generated by a language model directly: every figure in Sections 2 and 3 is produced by the code named in Section 8 and is reproducible from the committed artifacts, and the clinical figures are transcribed from the publications cited for them. Every literature identifier in Section 10 was checked against a retrieved bibliographic record held in this repository, and any identifier that could not be so anchored was removed rather than retained. The author takes full responsibility for all content, including for the correctness of the code and for the interpretation of the results.

Data and code availability. All code and all artifacts underlying every figure are in the public repository that accompanies this manuscript, under the paths given in Section 8. No restricted data were used.

Competing interests. The author declares no financial competing interests: he holds no position, equity, consultancy or patent relating to any gene, sequence, peptide or agent named here. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Funding. This work received no external funding and was self-funded by the author. No funder had any role in the analyses, the interpretation of the results, or the decision to publish.

Ethics. No human subjects, human material or animals were involved. No patient data were used; the clinical figures quoted are published or publicly presented aggregate results.

Not clinical guidance. Nothing in this manuscript is medical advice, and nothing in it is evidence that any agent or combination is safe or effective in extraskeletal myxoid chondrosarcoma. No peptide reported here has been synthesised, formulated, or tested in any cell, tissue or animal by anyone, and none may be administered to any person. The agents named in Section 4 are discussed as a research hypothesis and their use outside a clinical trial is not supported by anything in this paper.

10. References

Every identifier below is transcribed from a bibliographic record — either one held in this repository's literature and registry files, or one retrieved from Europe PMC or Crossref by the verification workflow that accompanies this work. None is written from recollection.

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and the reactive receptors mapped against fusion and wild-type sequence on patient-matched HLA. Cited for endogenous processing and presentation of a fusion breakpoint and for the restrictions recovered, and not as evidence of efficacy or safety in any disease. Retrieval record `research/literature/fusion-breakpoint-sweep-2026-08-24.json`.

Data sources cited as resources rather than as publications. Transcript structures are Ensembl records for *EWSR1* and *NR4A3*, retrieved and cached as committed inputs. The proteome searched in Section B5 is UniProtKB reference proteome UP000005640, reviewed entries with isoforms included, 42,547 sequences and 24,513,032 residues at retrieval, fetched from the UniProt REST stream; the isoform that carries the four colliding peptides is Q92570-3.

Appendix A. Superseded figures

The following values were reported before the 2026-08-07 coordinate-system correction and must not be quoted. They are retained so that earlier drafts and derived documents can be identified.

Quantity	Superseded value	Current value
In-frame junctions	7	5
Distinct predicted binders	26	11
EWSR1 exon 7 junction, presenting alleles	A*11:01 and B*08:01	B*15:01 on 10 alleles; B*15:01 and A*30:02 on 34
EWSR1 exon 7 junction coverage	29.7%	8.5% on 10 alleles, 12.3% on 34
Any-strong-binder coverage	58.0%	27.4% on 10 alleles, 30.4% on 34
Regional range	36% to 79%	1.4% to 60.4%
Broad-panel presenting alleles	20 of 34	4 of 34
Broad-panel coverage ceiling	84.5%	30.4%, and not a ceiling
Combined CD8 and CD4 coverage	16.5%	1.8%
Candidate minimal synthetic long peptide	27 residues, both arms	15 residues, both arms
Class II predicted binders at the *EWSR1* exon 7 junction	9, of which 4 strong	44 on 23 alleles, of which 1 strong

The superseded values arose from a model that concatenated coding sequences and thereby discarded the acceptor exon's retained 5' untranslated region. The corrected junction set is disjoint from the superseded one, so the earlier peptide identifiers do not appear in the current artifacts.

Appendix B. Statements withdrawn by the first adversarial review of this manuscript

This manuscript was reviewed on 2026-08-22 by five independent blind readers against a pinned commit, each with a different lens, none able to see the others. The statements below stood in the version they read and do not stand now. They are recorded because a reader who met an earlier copy, or a derived document that quoted one, needs to be able to identify what changed and why — and because a correction that leaves no trace is indistinguishable from a claim that was never made.

Withdrawn statement	Why it does not stand
"presented on HLA-B*15:01 alone", stated without naming a panel	True of the ten-allele screen only. The 34-allele screen finds the same lead peptide strong on HLA-A*30:02, and the junction's coverage is then 12.3% rather than 8.5%.
The 30.4% figure described as a "ceiling"	Sections 5 and 6 of the same manuscript said extending the panel could only raise it. It is a union over four alleles at one threshold, and moving that threshold to 0.37 takes it to zero.
"It does not attain 50% at any panel size" — withdrawn	No search over panel sizes was performed; the panel is fixed at 34 and the scanned variable is the number of presenting alleles. In Northern Europe two alleles reach 52.8%.
Wilson 95% confidence intervals on every coverage figure	They were computed by pooling every reference population into one binomial. The same records show the frequency of one pooled allele ranging from 0 to 0.40 between those populations, so the model the interval assumes is refuted by its own input.
"the combined CD8 and CD4 figure is null rather than unreported. That is a result" — withdrawn	The instrument's class II branch never evaluates when no allele qualifies, so the empty field records "not computed". The instrument's own note calls the class II coverage it would compute a floor that untested alleles could only raise.
"The one substantive reasoning correction this work offers …", and the symmetry it asserted	The quoted exclusions are this author's own route-ledger notes, not positions in the literature, and the vaccine's own entry in that ledger is parked on immunogenicity rather than on absent priming. The claim is narrowed in B6 and the Abstract accordingly.
The dated capability bands of the former Section 3.1 (2027H2 / 2029 / 2032)	Withdrawn entire. They were self-described as the weakest material in the paper, and a forecast a reader cannot check does not become checkable by being tabulated.
"regenerated in continuous integration", of the artifacts in Section 8	The workflow is dispatched by hand, its steps do not fail the run when a generator fails, and it writes to a cache branch. Section 8 now says what is actually true.
Reference 8 quoted as "correlated with disease outcome"	The study's title names immune response in placenta and colorectal cancer, which is the tissue setting it examined.
References 10 and 11 as "[citation to verify]"	Both were resolvable in this repository at no cost and are now written out. The one reference that genuinely has no record — the class II predictor — says so in its own entry instead.
"24 patients across nine centres … gives a realistic accrual rate"	Unfiltered by this paper's own eligibility criteria. Applying them gives roughly 0.3 to 1.4 eligible patients a year.

Appendix C. Statements withdrawn after the first adversarial review

The review recorded in Appendix B was not the last correction. The statements below stood after it and do not stand now, and they are recorded on the same principle: a correction that leaves no trace is indistinguishable from a claim that was never made.

Withdrawn statement	Why it does not stand
B8, "rather than from published EMC-specific immune profiling"	Such profiling was published on 2026-07-13, four weeks before this manuscript's date, on the twelve-tumour cohort described in B8's own revision.
B8, HLA class I expression status "is not known for this disease"	The antigen-presentation precondition — β2-microglobulin, the class I heavy chains, the peptide transporters, tapasin, NLRC5, the immunoproteasome subunits and ERAP1 — had been read on both EMC array platforms on 2026-08-09, in this author's own artifacts, before the sentence was written.
B6 and B8 read together, that whether EMC is immunologically quiet was unmeasured here	It is now measured in the two archival cohorts, and the direction is that EMC is quieter than the comparator sarcomas on the interferon axis, the myeloid compartment and class II presentation. That is a cohort contrast against other sarcomas and not a statement that the disease is cold; the distinction is stated in B8.
Section 1, "no published EMC immune profiling" listed among the limits of access	The same statement as B8's, in a second place, and false for the same reason. It is the one-of-a-pair defect: correcting B8 alone would have left the claim standing where a reader meets it first.
Section 7, "The characterisations of EMC as cold and as immune-excluded rest on inference rather than on published EMC-specific immune profiling"	Third home of the same claim. The exclusion half still rests on inference; the quiet half no longer does.
Section 7, "The class II panel is three DR alleles with no DP or DQ, so its negative bounds a narrow question"	Superseded by the panel widening reported in B4 and in Appendix A, and stale in this section since that widening. Section 7 was contradicting B4 inside the same document: the panel is 23 alleles and the arm is not negative. This was not found by any adversarial round; it is a wording that no numeric guard reads.
B7, "No EMC-specific expression evidence for the pathway is cited, because none is known to the author"	The chondroitin-sulfate biosynthetic and sulfation machinery had been read on both EMC array platforms in this author's own artifacts before the sentence was written. The reading is reported in B7 and does not support the analogy the sentence was protecting, which is a different correction from the one the sentence claimed to make.
B2, that the reports available "do not close this for any fusion", because they measure responses to peptides that were administered	A study of 34 paediatric acute leukaemias recovered fusion-breakpoint-specific T-cell receptors from co-culture with autologous blasts, with no peptide administered, and mapped their restrictions on patient-matched HLA. Endogenous processing and presentation of a fusion breakpoint is shown; presentation on an EMC cell, and abundance anywhere, are not.
B4, that the class II arm is the lesser of the two	Two of three fusion-specific restrictions recovered from endogenously presented breakpoints were class II, and the one reported de novo vaccine response was CD4+. Three observations are a pattern to test, and B4 now asserts neither ranking; what remains ordered is the accuracy of the two predictors.
Section 6, that no report on another fusion "measured presentation at all"	The same study measures processing and presentation, though not abundance. The bullet now separates the two, because step 3 of Section 6.1 turns on abundance.
The B6 and B7 rows of Section 3, cost to move "not movable by any computation"	True of each limit and false of the evidence bearing on it: a computation over public archival series moved what is known about both. Those cells now name what the remaining move costs, which is spatial profiling on tissue.
B3, that a difference at T-cell-receptor contact positions rather than at anchors is "the configuration in which cross-recognition is most plausible and in which central tolerance is most likely to have acted"	Inverted, and it contradicted the next sentence of its own section. B3 defines anchor positions as the ones affecting binding and contact positions as the ones affecting recognition, and then calls an anchor-only neighbour "the worst case: an identical TCR-facing surface distinguished only by residues the T cell cannot see". A difference the receptor reads is what leaves a repertoire possible, so the ordering runs the other way. The clause had been carried onward into falsifier 3 of `shared-vs-individualized-neoantigen-evidence.md`, which is corrected in the same direction.
B3 and the Abstract, "no binder in the screen has a near-self neighbour differing only at anchors", stated without the convention it depends on	The count is zero under the general class I convention of position 2 and the C-terminus and stays zero when position 3 is added, which is the caveat this paper had raised against itself. Counting position 1 as an anchor makes it six of the 11 binders, all six against the same *NR4A3* isoform, and no allele-specific motif for the six restricting alleles is held in this work. Both statements now name the convention and the six.
Section 7, "with no decoy control and no null expectation", and the calls "reported as what the screen returned rather than as an enrichment over chance"	A decoy null now exists and the screen sits below it. Length-matched random peptides from the same reviewed human proteome pass at 0.644% of 59,160 peptide-allele tests, and a random set of the screen's own size presents on a median of 23 of the 34 panel alleles against this screen's 4. What was disclosed as a missing control is now a measured deviation from chance in the negative direction, reported in Section 7. It bears on the screen and not on whether any of the four strong calls is real.